

GSP 216 Final Project Report

Introduction

The main purpose of my project is to calculate the amount of spring snow cover seen around the Lake Tahoe Heavenly Lake Ski Resort between the years of 1995 and 2020. Snow cover was chosen as my focus topic due to the inherent interest I have in it both as a recreational outdoors person and an aspiring environmental professional. I hope to gain further insight into how snow cover has been changing in popular recreational ski and snowboard areas in the form of exact data reflecting snow cover in the spring ski month of April. As an ESM major with a specialization in natural resource recreation, I believe data such as this can be instrumental in creating effective policies for future management of these increasingly popular destinations. The project will also serve to demonstrate some of the knowledge and technique learned in the class over the semester.

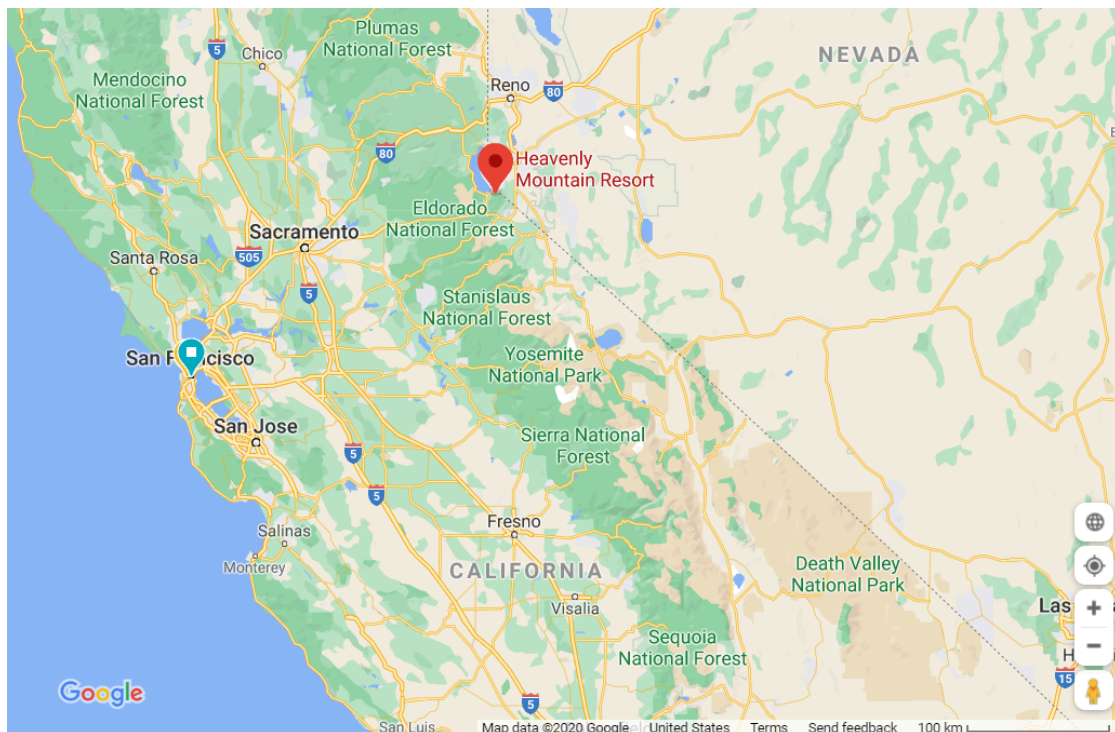


Figure 1: Heavenly Mountain Resort, Tahoe, CA

In recent years, California has seen many different consequences of a changing climate. Wildfires are burning at rates never seen before, drought has been negatively impacting our agricultural sector for years, and snowpack has seen much more volatile measurements (Sun, 2016). And although the effects of these events have yet to cause any widespread and catastrophic changes to our economy and society, it is very possible that if we refuse to change our habits, these effects will one by one begin to encroach on our lives. The total potential economic loss of a heavily impacted snowpack has been estimated to be in the trillions of dollars (Sturm, 2017), and keeping a keen eye on many of the locations which may be more sensitive to this potential economic loss is something that I believe will be very important for the creation of effective adaptations to management policy.

Methods

The data to be analyzed was downloaded exclusively through the USGS EarthExplorer tool. Since the dates of the data that were to be analyzed spanned multiple generations of satellite technology, a variety of sensor types were used for the project. Specifically LANDSAT 8 for data after 2010, LANDSAT 7 for data between 2000 and 2010, and LANDSAT 5 for anything before 2000. All of the sensors utilize a 30x30 meter spatial resolution, and the images used for analysis were obtained from each sensor's bands corresponding to the 0.53 - 0.61 micrometer wavelength and the 1.55 - 1.75 micrometer wavelength. Landsat 8 uses bands 3 and 6, Landsat 7 uses bands 2 and 5, which are also the same band numbers for Landsat 5. These bands measure the visible green and shortwave infrared wavelengths, both of which are used in the calculation of the Normalized Difference Snow Index (NDSI). The Raster Calculator tool in the spatial analyst toolbox was used to perform the NDSI calculation, which is
$$\text{NDSI} = (\text{Band for Visible Green} - \text{Band for SWIR}) / (\text{Band for Visible Green} + \text{Band for SWIR}).$$
 The resulting file was then classified manually into 5 sections, corresponding to rock, dirt, melting snow, thin snow, and thick snow using classification intervals of 0, 0.2, 0.53, 0.7, and 1. This resulting NDSI image was then used to calculate the pixel area of each classification using the zonal histogram tool. Color schemes were applied to the image to allow for ease of legibility. Since the full extent of the images encompassed areas

outside of the focus zone, a shapefile containing only the resort and its ski runs was created to be used as a zonal feature input shapefile.

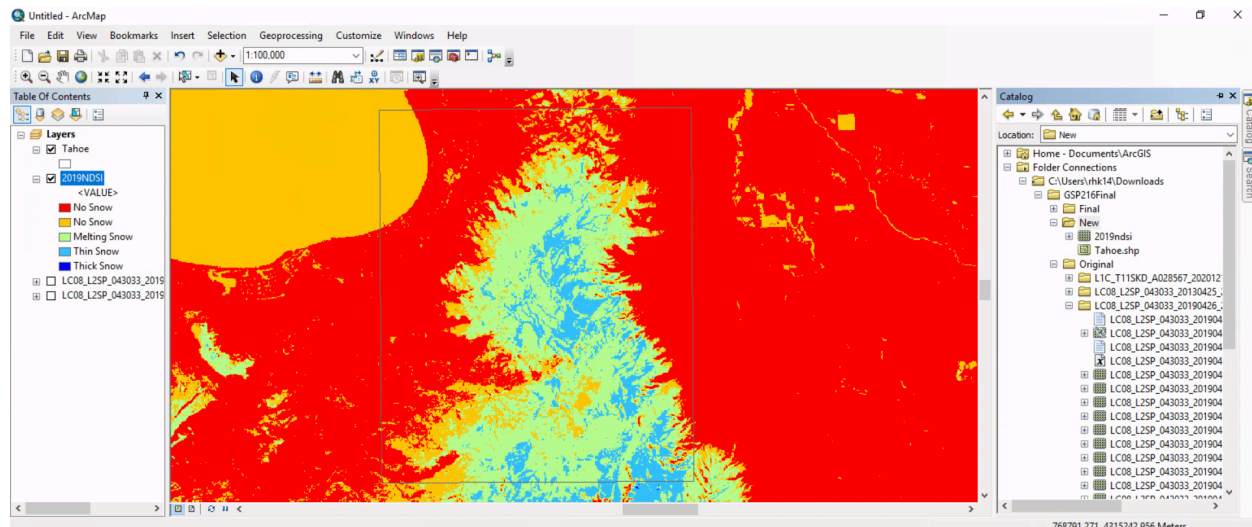


Figure 1: An NDIS like the one above was created for each of the indicated years.

Results

5 total NDSI images were produced of the Tahoe Heavenly Ski Resort area for the years 1995, 2000, 2008, 2013, and 2019. From these images, 5 snow coverage tables were created using the zonal histogram tool. The total area for classifications 3, 4, and 5 (Melting snow, thin snow, and thick snow) were summed up to provide a single variable corresponding to the area of snow cover for each year for the specified location of Heavenly Lake Ski Resort.

Table 1: Pixels of types of snow detected for each year using NDSI

Year/Snow	1995	2000	2008	2013	2019
Melting snow (Pixels)	44745	21654	28540	21760	43119
Thin snow (Pixels)	19126	275	218	2193	11867
Thick snow (Pixels)	2239	10873	10920	0	0
Total Pixels	66110	32802	39678	23953	54986
Total Area (Hectares)	5949.9	2952.18	3571.02	2155.77	4948.74
Melting snow (Hectares)	4027.05	1948.86	2568.6	1958.4	3880.71
Thin snow (Hectares)	1721.34	24.75	19.62	197.37	1068.03
Thick snow (Hectares)	201.51	978.57	982.8	0	0

The total number of pixels was transcribed from the table created in ArcMaps using the Zonal Histogram tool into Microsoft Excel. These pixels correspond to sensors which all have a 30x30 meter spatial resolution, so a simple formula $(X * 30 * 30) / 10000$ was used to convert the pixels of each type of snow found into hectares of snow.

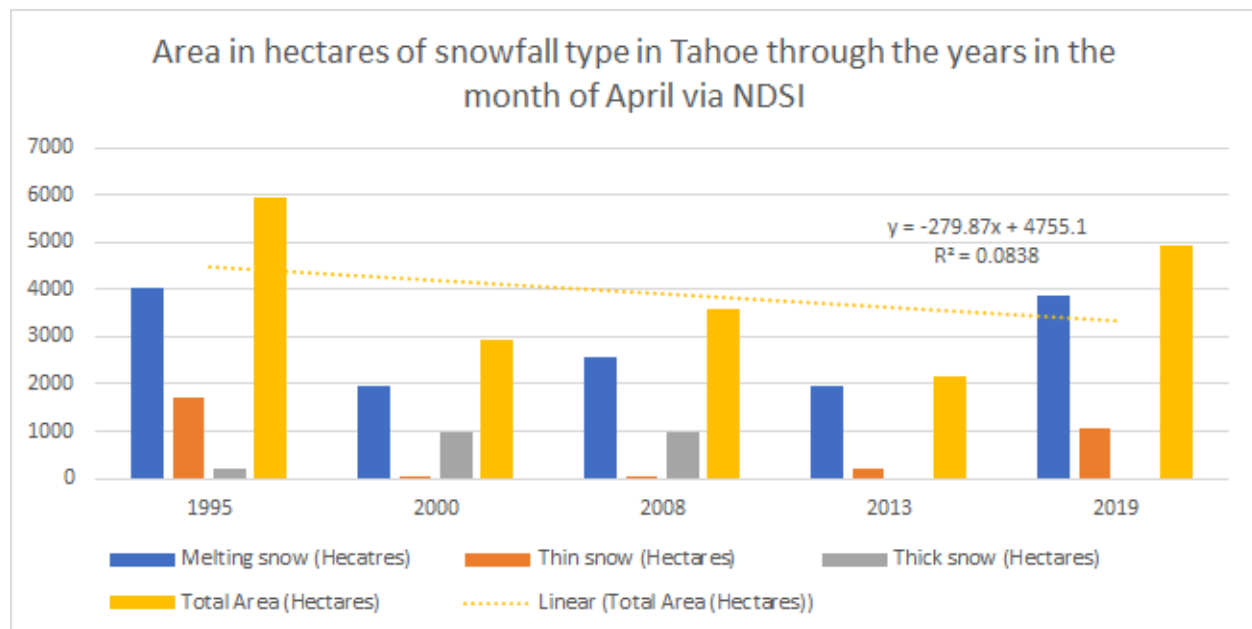


Figure 1: Total area of each snow cover type in hectares graphed over year observed

The graph of the total area of snow covered each year produced a trendline with a slope of -279.87 and an R squared value of 0.08.

Discussion

The values obtained from the analysis are in line with the hypothesized theory of less unmelted snowfall each year in the month of April. However it should be noted that the low number of data points makes drawing a statistically sound conclusion more difficult. Further analysis is possible, as all required data is already online and available for download. The negative slope displayed by the trendline of the total measured snowfall area suggests a slow but steady decline in whatever snow is remaining in

Heavenly Lake Tahoe by the month of April. The snowfall during periods of drought in the mid 2000's also shows a visibly lower area of unmelted snow as compared to the late 90's and more recently in 2019. We can also observe that even though the total area of snow has rebounded a bit since the severe drought period, we are still seeing reduced amounts of remaining thick snowfall.

Conclusion

I believe I chose an interesting subject and am satisfied with the analysis conducted. It is an exciting feeling to be able to directly calculate statistics that even in this age of informational availability would otherwise be difficult or impossible to find. The fact that there are so many different spectral bands that can be used in so many different ways has been something I have been slowly comprehending over the semester, and this final project was an excellent way to cement this knowledge to memory. I believe there is merit in studying the locations that are close to my heart in ways that may not be so easy to discover only through the help of google. Snowfall, vegetation on trails, erosion and burn areas, the list of subjects that can be further explored are endless. The fact that all of this data is available at the click of a button is icing on the cake. I believe that with my intended career path in the natural resource recreation profession, the abilities learned in this class will continue to benefit me for many years to come.

Bibliography

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